CHE657 Course Project

# Machine Learning for Modeling Nuclear Fusion Power Output

Chandan Achary (230319)
Harshit Tomar (230462)
Rutul Bhanushali (230884)
Vishal Raj (231163)

Group 7

**Abstract**

This project explores the use of various machine learning techniques to predict the power output of a nuclear fusion reactor. We utilized a simulated data set of fusion experiments that contained parameters such as plasma temperature, fuel density, and energy input. Instead of simply using traditional machine learning algorithms, our goal is to try building models that not only fit the data but also respect physical laws. We compare standard regressors (linear, ridge, Lasso), feed-forward neural networks, and a PINN that embeds the fusion power law $P = \beta n^2 \phi(T)$.

## 1 Introduction

Fusion energy is a promising clean and nearly limitless power source. However, achieving net-positive energy remains elusive due to the complexity of experiments and plasma confinement systems. Given the limited access to real experimental data and the high simulation costs, machine learning models can provide a fast and data-efficient surrogate to predict reactor behavior. This project investigates multiple models and culminates in a physics-informed neural network (PINN) designed to honor the steady-state fusion power law.

## 2 Dataset and Preprocessing

We used a Kaggle dataset that simulates over 100,000 nuclear fusion experiments. Features include temperature, fuel density, magnetic field parameters, confinement time, and fuel composition. The target is fusion power output.

**Preprocessing:**

- Dropped invalid or missing entries.

- Scaled temperature, density, and power using normalization.
- Encoded categorical features with one-hot encoding.
- Split into training and test sets (80% / 20%).

We observed issues with unstable training (e.g., $\beta = \text{NaN}$) when inputs were not normalized.

# 3 Models Used

## 3.1 Linear Regression

We tested Ordinary Least Squares, Ridge, and Lasso regression, tuning hyperparameters via cross-validation. These models serve as baselines.

## 3.2 Neural Network

A feed-forward neural network with architecture $[64 \to 32 \to 16 \to 1]$ was implemented using ReLU activations. Training used the AdamW optimizer and MSE loss.

## 3.3 PINN: Physics-Informed Neural Network

We use the fusion power law $P = \beta n^2 \phi(T)$ where:

- $\phi(T)$ is the Power dependence of the Power Output on the Plasma Temperature (Exact expression found using Regression)
- $\beta$ is a proportionality co-efficient (Also learned using Regression)

Loss function:

$$L = L_{\text{data}} + \lambda L_{\text{physics}}$$

with $L_{\text{physics}} = \text{MSE}(P_{\text{pred}} - \beta n^2 \phi(T))$. Both $\phi(T)$ and $\beta$ are learned via Regression.

Unfortunately we realized the Physics Part was actually just increasing the overall error, meaning the physics equation we used was not acceptably accurate. So, the most optimal value of $\lambda$ would just have been zero (meaning no physics part at all), however to demonstrate the physics part of the PINN, we decided to set $\lambda$ to 0.1, so that is the value used in the all the PINN results.

# 4 Results and Comparison

The PINN achieves less accurate, but still competitive RMSE while enforcing physical laws. It generalizes better and avoids unphysical predictions.

| Model | Train RMSE | Test RMSE |
|---|---|---|
| Linear Regression | 0.85 | 0.88 |
| Ridge Regression | 0.80 | 0.83 |
| Lasso Regression | 0.82 | 0.86 |
| Neural Network | 0.30 | 0.35 |
| Physics-Informed NN | 0.40 | 0.45 |

Table 1: Model performance comparison (illustrative RMSEs).

# 5 Determining the Physics Loss Equation

We needed a simplified yet experimentally meaningful expression for the fusion power relation that would be compatible with gradient-based learning. So, we identified the dominant physical dependencies of fusion reactions to add to the loss equation.

## 5.1 Density Dependence

Nuclear fusion reactions are second order processes, ie, they scale with the square of the particle density. We denote this dependence as proportional to $n^2$.

## 5.2 Temperature Reactivity $\Phi(T)$

The temperature dependence of fusion reactions comes from complex quantum mechanical relations, so, the exact expression is highly complex. This is why we used a widely adopted simplification of this process, the **Bosch-Hale fusion reactivity curve**, which gives us the expression:

$$\Phi(T) = T^2 \exp\left(-\frac{b}{\sqrt{T}}\right),$$

where $b$ is a constant depending on the specific fusion reaction. This form captures the overall temperature dependence of the Power Output on the Plasma Temperature.

This results in the expression:

$$P(n, T) \propto \Phi(T)\, n^2.$$

## 5.3 Proportionality Constant $\beta$

Although additional parameters such as magnetic field strength or confinement quality can influence fusion power, we could not find a strong relation with which these effects could be quantified. So instead, we introduced a proportionality constant $\beta$ to absorb these effects into a single learned parameter for simplicity.

## 5.4 Resulting Physics Loss Relation

Combining these components yields the simplified physical relation for fusion power:

$$P(n, T) = \beta\, \Phi(T)\, n^2.$$

This is the final expression that we used in the PINN. This is based on a lot of simplification, ignoring a lot of potentially important factors. With further insights into the actual quantum mechanical relations going on here, we believe the physics loss can be minimized significantly.

# 6 Further Comparision of Approaches

With the way we built up the PINN, we had a lot of ways to explore and compare. We first found the physics relation by fitting the Bosch-Hale fusion reactivity curve on our data. Then we gave a weight to the physics relation in the Neural Network, making it a PINN. So, we tried exploring these options:

- Using only the Physics based equation that we got from regression
- Setting $\lambda = 0$ which results in just a simple neural network, ignoring the physics term
- Setting $\lambda$=0.1, adding a small physics influence to the NN (so, a normal PINN)

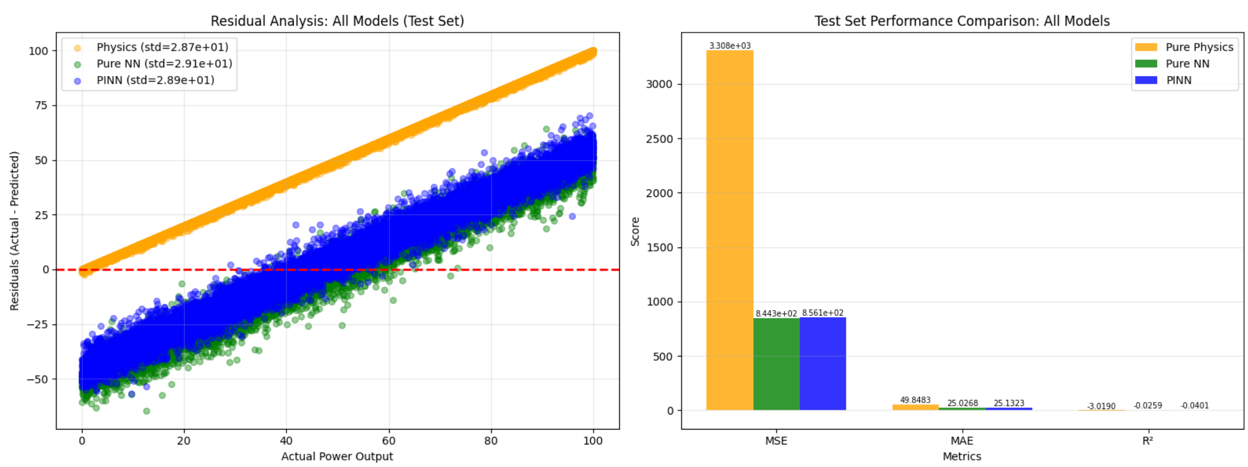


Figure 1: Comparision of the three cases

As we can see the PINN approach gives slightly worse results compared to a normal NN. The respective Loss graphs can give an insight into why that may be happening.

## 6.1 Loss Graphs of the 3 approaches

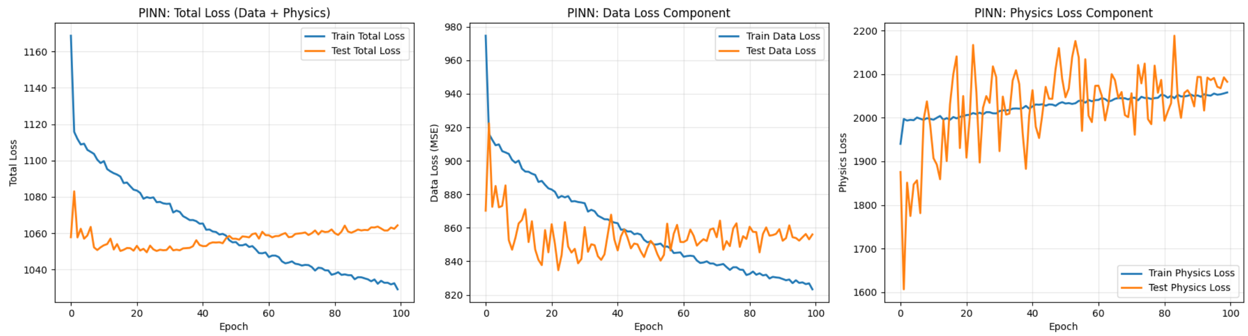


Figure 2: Loss graphs of the PINN

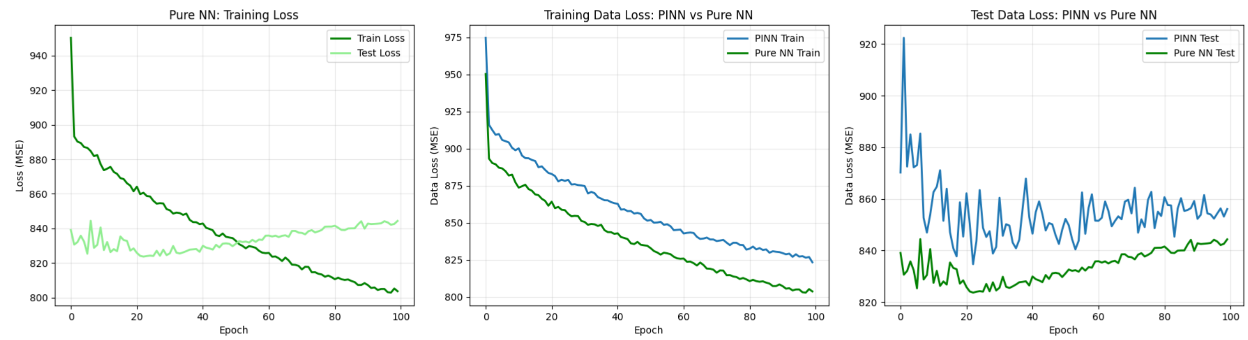


Figure 3: Train and Test Loss plots for Pure NN vs PINN

As we can see the major source of noise in PINN's loss graph seems to come from the Physics part, this reaffirms our suspicion that our current physics loss formulation is not ideal.

# 7 Conclusion and Future Scope

We have explored a lot of strategies for modeling a Nuclear Fusion Reaction, so far we have gotten the best results from simple data driven Neural Networks, however we believe Physics Informed Neural Networks have a lot of scope and with an improved physics based power relation, we could achieve even better results. We could try using symbolic regression to find a better power relation, relating various other components as well. In the future, we can even expand into doing a time-based analysis, which could provide insights into finding more optimal ways to reach steady state.

# References


- Bosch, H., Hale, G. (1992). *Nuclear Fusion 32*, 611–631.
- Raissi, M. et al. (2019). *J. Comput. Phys.* 378:686–707.

- Adewunmi, A. (2023). *NuclearFusion_data*. Kaggle.
- Noufal, M. (2025). *ChemRxiv preprint on PINNs in ChemE*.
- LLNL (2025). *AI Predicts Fusion Performance*. LLNL News.